Elements of Matrix Algebra

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Introduction

The material covered here may be found in numerous appendices of econometrics textbooks. We will only give definitions and results relevant to econometrics (you will be asked to verify some in the exercises). For more, see, e.g.,

- Rao and Rao (1998) *Matrix Algebra and Its Applications to Statistics and Econometrics*. World Scientific, 1st ed.
  - Has the necessary background on linear algebra.
- Magnus and Neudecker (2019) *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Wiley, 3rd ed.
  - The title summarizes it aptly.
- Lütkepohl (1996) *Handbook of Matrices*. Wiley, 3rd ed.
  - Has a lot more results for ready reference.

Definitions, Notation, Terminology

Definition 0.1: Matrix.

A $(m \times n)$ **matrix** $\boldsymbol{A}$ is an array of numbers:

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

Alternative notation: $\boldsymbol{A}\ (m \times n)$, $\boldsymbol{A} = [a_{ij}]\ (m \times n)$.

m and n are positive integers denoting the **row dimension** and the **column dimension**, respectively, $(m \times n)$ is the **dimension** or order of $\boldsymbol{A}$.

For our purposes, the a_{ij}, $i = 1, \ldots, m$, $j = 1, \ldots, n$, are elements of $\mathbb{R}$.

As such, $\boldsymbol{A}$ can be viewed as a linear mapping $\boldsymbol{A} : \mathbb{R}^n \to \mathbb{R}^m$.

Definition 0.2: Block Matrix.

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{A}_{11} & \cdots & \boldsymbol{A}_{1n} \\ \vdots & \ddots & \vdots \\ \boldsymbol{A}_{m1} & \cdots & \boldsymbol{A}_{mn} \end{pmatrix}$$

Here, the $\boldsymbol{A}_{ij}$ are $(m_i \times n_j)$ submatrices of $\boldsymbol{A}$. A matrix written in terms of submatrices rather than individual elements is often called a **partitioned matrix** or a **block matrix**.

Definition 0.3: Special Matrices and Elements.

A matrix $\boldsymbol{A}$ $(m \times m)$ having the same number of rows and columns is a **square** or **quadratic matrix**.

The elements a_{ii}, $i = 1, \ldots, m$, constitute its **principal diagonal**.

If $a_{ii} = 1$ for $i = 1, \ldots, m$, and $a_{ij} = 0$ for $i \neq j$ is an $(m \times m)$ **identity matrix**, denoted $\boldsymbol{I}_m$.

A $(1 \times n)$ matrix $\boldsymbol{a}$ is an n-dimensional **row vector** or $(1 \times n)$ vector.

An $(m \times 1)$ matrix $\boldsymbol{b}$ is an m-dimensional **column vector**.

Definition 0.4: Trace and Transpose.

The **trace** of $\boldsymbol{A}$ is defined as

$$\mathrm{tr}(\boldsymbol{A}) = \sum_{i=1}^{m} a_{ii}$$

The **transpose** of $\boldsymbol{A}$ $(m \times n)$ is defined as

$$\boldsymbol{A}^\top = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}^\top = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}$$

$\boldsymbol{A}^\top$ is $(n \times m)$. That is, the rows of $\boldsymbol{A}$ are the columns of $\boldsymbol{A}^\top$.

Definition 0.5: Determinant.

The **determinant** of $\boldsymbol{A}$ $(m \times m)$

$$\begin{aligned} \det(\boldsymbol{A}) = |\boldsymbol{A}| &= a_{i1}\mathrm{cof}(a_{i1}) + \ldots + a_{im}\mathrm{cof}(a_{im}) \\ &= a_{1j}\mathrm{cof}(a_{1j}) + \ldots + a_{mj}\mathrm{cof}(a_{mj}), \end{aligned}$$

where $\mathrm{cof}(a_{ij}) = (-1)^{i+j}\mathrm{minor}(a_{ij})$ and

$$\mathrm{minor}(a_{ij}) = \begin{vmatrix} a_{1,1} & \cdots & a_{1,j-1} & a_{1,j+1} & \cdots & a_{1,m} \\ \vdots & & \vdots & \vdots & & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,j-1} & a_{i-1,j+1} & \cdots & a_{i-1,m} \\ a_{i+1,1} & \cdots & a_{i+1,j-1} & a_{i+1,j+1} & \cdots & a_{i+1,m} \\ \vdots & & \vdots & \vdots & & \vdots \\ a_{m,1} & \cdots & a_{m,j-1} & a_{m,j+1} & \cdots & a_{m,m} \end{vmatrix}$$

Definition 0.6: Inverses.

For $\boldsymbol{A}$ $(m \times m)$ with $\det(\boldsymbol{A}) \neq 0$, the **inverse** of $\boldsymbol{A}$ is the unique $(m \times m)$ matrix $\boldsymbol{A}^{-1}$ satisfying $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_m$.

An $(n \times m)$ matrix $\boldsymbol{A}^{-}$ is a **generalized inverse** of the $(m \times n)$ matrix $\boldsymbol{A}$ if it satisfies $\boldsymbol{A}\boldsymbol{A}^{-}\boldsymbol{A} = \boldsymbol{A}$. $\boldsymbol{A}^{-}$ is generally not unique.

The $(n \times m)$ matrix $\boldsymbol{A}^{+}$ is the **Moore–Penrose (generalized) inverse** of the $(m \times n)$ matrix $\boldsymbol{A}$ if it satisfies

- $\boldsymbol{A}\boldsymbol{A}^{+}\boldsymbol{A} = \boldsymbol{A}$,
- $\boldsymbol{A}^{+}\boldsymbol{A}\boldsymbol{A}^{+} = \boldsymbol{A}^{+}$,
- $(\boldsymbol{A}\boldsymbol{A}^{+})^\top = \boldsymbol{A}\boldsymbol{A}^{+}$,
- $(\boldsymbol{A}^{+}\boldsymbol{A})^\top = \boldsymbol{A}^{+}\boldsymbol{A}$.

$\boldsymbol{A}^{+}$ exists and is unique.

Definition 0.7: Certain Matrix Products.

Power of a matrix $\boldsymbol{A}$ $(m \times m)$:

$$\boldsymbol{A}^i = \begin{cases} \prod_{j=1}^{i} \boldsymbol{A} = \underbrace{\boldsymbol{A} \times \cdots \times \boldsymbol{A}}_{i \text{ times}} & \text{for positive integers, } i \in \mathbb{Z}^+ \\ \boldsymbol{I}_m & \text{for } i = 0 \\ (\prod_{j=1}^{-i} \boldsymbol{A})^{-1} & \text{for } i \in \mathbb{Z}^-, \text{if } \det(\boldsymbol{A}) \neq 0 \end{cases}$$

The $(m \times m)$ matrix $\boldsymbol{A}^{1/2}$ is a **square root** of the $(m \times m)$ matrix $\boldsymbol{A}$ if $\boldsymbol{A}^{1/2}\boldsymbol{A}^{1/2} = \boldsymbol{A}$.

Definition 0.8: Vectorization.

The **vectorization** of $\boldsymbol{A}$ $(m \times n)$ is

$$\operatorname{vec}(\boldsymbol{A}) = \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \\ a_{12} \\ \vdots \\ a_{m2} \\ \vdots \\ a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \quad (mn \times 1)$$

That is, vec stacks the columns of $\boldsymbol{A}$ in a column vector.

Definition 0.9: Some Matrix Properties.

1. An $(m \times m)$ matrix with $\boldsymbol{A} = \boldsymbol{A}^\top$ is **symmetric**.
2. An $(m \times m)$ matrix $\boldsymbol{A}$ is **idempotent** if $\boldsymbol{A}^2 = \boldsymbol{A}$.
3. An $(m \times m)$ matrix $\boldsymbol{A}$ is **nonsingular** or **invertible** or **regular** if $\det(\boldsymbol{A}) \neq 0$ and thus $\boldsymbol{A}^{-1}$ exists.
4. An $(m \times m)$ matrix $\boldsymbol{A}$ is **orthogonal** if $\boldsymbol{A}$ is nonsingular and $\boldsymbol{A}^\top = \boldsymbol{A}^{-1}$.

Definition 0.10: Definiteness.

A symmetric $(m \times m)$ matrix $\boldsymbol{A}$ is

- **positive definite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ for all $(m \times 1)$ vectors $\boldsymbol{x} \neq \boldsymbol{0}$
- **positive semidefinite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geqslant 0$ for all $(m \times 1)$ vectors $\boldsymbol{x}$
- **negative definite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} < 0$ for all $(m \times 1)$ vectors $\boldsymbol{x} \neq \boldsymbol{0}$
- **negative semidefinite** if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \leqslant 0$ for all $(m \times 1)$ vectors $\boldsymbol{x}$
- **indefinite** if there exist $(m \times 1)$ vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ such that $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ and $\boldsymbol{y}^\top \boldsymbol{A}\boldsymbol{y} < 0$.

Definition 0.11: Triangular Matrices.

An $(m \times m)$ matrix

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 0 \\ a_{m1} & \cdots & \cdots & a_{mm} \end{pmatrix}$$

is **lower triangular**.

A matrix $\boldsymbol{A}$ $(m \times m)$ that is both upper and lower triangular is a **diagonal matrix**, denoted $\boldsymbol{A} = \text{diag}(a_{11}, \ldots, a_{mm})$.

Further Concepts

Definition 0.12:

The m-dimensional row or column vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$ are **linearly independent** if, for scalars $c_1, \ldots, c_k$, $c_1\boldsymbol{x}_1 + \ldots + c_k\boldsymbol{x}_k = \boldsymbol{0}$ implies $c_1, \ldots, c_k = 0$.

Definition 0.13:

The **rank** of $\boldsymbol{A}$ $(m \times n)$, denoted $\mathrm{rk}(\boldsymbol{A})$, is the maximum number of linearly independent rows or columns of $\boldsymbol{A}$.

Definition 0.14:

Given a symmetric $(m \times m)$ matrix $\boldsymbol{A}$, the function $Q : \mathbb{R}^m \to \mathbb{R}$ defined by $Q(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}$ is called a **quadratic form**.

Definition 0.15:

The polynomial in $\boldsymbol{A}$ given by $\det(\lambda \boldsymbol{I}_m - \boldsymbol{A})$ is the **characteristic polynomial** of the $(m \times m)$ matrix $\boldsymbol{A}$.

Definition 0.16: Eigenvalues and Eigenvectors.

The **eigenvalues** of an $(m \times m)$ matrix $\boldsymbol{A}$ are the roots of the characteristic polynomial of $\boldsymbol{A}$.

(Note that eigenvalues can be complex depending on the polynomial.)

An $(m \times 1)$ vector $\boldsymbol{v} \neq \boldsymbol{0}$ satisfying $\boldsymbol{Av} = \lambda \boldsymbol{v}$, where λ is an eigenvalue of $\boldsymbol{A}$, is an **eigenvector** corresponding to the eigenvalue λ.

Kronecker Product

Definition 0.17:

The **Kronecker product** of $\boldsymbol{A}$ $(m \times n)$ and $\boldsymbol{B}$ $(p \times q)$ is defined as the $(mp \times nq)$ matrix

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{pmatrix} a_{11}\boldsymbol{B} & \cdots & a_{1n}\boldsymbol{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\boldsymbol{B} & \cdots & a_{mn}\boldsymbol{B} \end{pmatrix}$$

Operations Relating Matrices

Definition 0.18:

1. **Addition:** $\boldsymbol{A}\,(m \times n),\ \boldsymbol{B}\,(m \times n) : \boldsymbol{A} + \boldsymbol{B} = [a_{ij} + b_{ij}]\,(m \times n)$
2. $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}\,(m \times n) : (\boldsymbol{A} \pm \boldsymbol{B}) \pm \boldsymbol{C} = \boldsymbol{A} \pm (\boldsymbol{B} \pm \boldsymbol{C})$
3. **Multiplication by a scalar:** $\boldsymbol{A}\,(m \times n),\ c$ a scalar $c\boldsymbol{A} = [c \cdot a_{ij}]\,(m \times n)$
4. **Matrix multiplication** or matrix product: $\boldsymbol{A}\,(m \times n),\ \boldsymbol{B}\,(n \times p)$

$$\boldsymbol{AB}_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \ \ \forall\ i = 1, \ldots, m;\ j = 1, \ldots, p\ (m \times p)$$

   If the blocks are conformable, multiplication of partitioned matrices proceeds analogously.

Rules

Proposition 0.19:

1. $\boldsymbol{A}\ (m \times n)$: $\boldsymbol{I}_m \boldsymbol{A} = \boldsymbol{A} \boldsymbol{I}_n = \boldsymbol{A}$
2. $\boldsymbol{A}\ (m \times n), \boldsymbol{B}, \boldsymbol{C}\ (n \times r)$: $\boldsymbol{A}(\boldsymbol{B} \pm \boldsymbol{C}) = \boldsymbol{A}\boldsymbol{B} \pm \boldsymbol{A}\boldsymbol{C}$
3. $\boldsymbol{A}\ (m \times n), \boldsymbol{B}, \boldsymbol{C}\ (p \times q)$: $\boldsymbol{A} \otimes (\boldsymbol{B} \pm \boldsymbol{C}) = \boldsymbol{A} \otimes \boldsymbol{B} \pm \boldsymbol{A} \otimes \boldsymbol{C}$
4. $\boldsymbol{A}, \boldsymbol{B}\ (m \times n)$: $(\boldsymbol{A} \pm \boldsymbol{B})^\top = \boldsymbol{A}^\top \pm \boldsymbol{B}^\top$
5. $\boldsymbol{A}, \boldsymbol{B}\ (m \times m)$: $\mathrm{tr}(\boldsymbol{A} \pm \boldsymbol{B}) = \mathrm{tr}(\boldsymbol{A}) \pm \mathrm{tr}(\boldsymbol{B})$
6. $\boldsymbol{A}, \boldsymbol{B}\ (m \times n)$: $\mathrm{rk}(\boldsymbol{A} \pm \boldsymbol{B}) \leqslant \mathrm{rk}(\boldsymbol{A}) + \mathrm{rk}(\boldsymbol{B})$
7. $\boldsymbol{A}\ (m \times n), \boldsymbol{B}\ (n \times p)$: $\mathrm{rk}(\boldsymbol{A}\boldsymbol{B}) \leqslant \min(\mathrm{rk}(\boldsymbol{A}), \mathrm{rk}(\boldsymbol{B}))$
8. $\boldsymbol{A}\ (m \times m)$: $\boldsymbol{A} + \boldsymbol{A}^\top$ is a symmetric $(m \times m)$ matrix

Rules (2)

Proposition 0.20:

1. $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC}) = \boldsymbol{ABC}$
2. $\boldsymbol{AB} \neq \boldsymbol{BA}$ in general
3. $\boldsymbol{A}\ (m \times n), \boldsymbol{B}\ (p \times q), \boldsymbol{C}\ (n \times r), \boldsymbol{D}\ (q \times s) : (\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D}) = \boldsymbol{AC} \otimes \boldsymbol{BD}$.
4. $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$
5. $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$
6. $\boldsymbol{A}\ (m \times n), \boldsymbol{B}\ (n \times m) : \operatorname{tr}(\boldsymbol{AB}) = \operatorname{tr}(\boldsymbol{BA})$
   This holds more generally for any cyclical permutation of matrices $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ such that the product remains defined.
7. $\boldsymbol{A}, \boldsymbol{B}\ (m \times m) : \det(\boldsymbol{AB}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$
8. $\boldsymbol{A}\ (m \times n) : \operatorname{rk}(\boldsymbol{A}) = \operatorname{rk}(\boldsymbol{A}^\top) = \operatorname{rk}(\boldsymbol{A}^\top \boldsymbol{A}) = \operatorname{rk}(\boldsymbol{AA}^\top)$
9. $\boldsymbol{A}\ (m \times n)$: $\boldsymbol{A}^\top \boldsymbol{A}$ and $\boldsymbol{AA}^\top$ are symmetric positive semidefinite.
   If, furthermore, $\operatorname{rk}(\boldsymbol{A}) = n$, then $\boldsymbol{A}^\top \boldsymbol{A}$ is positive definite.

Rules (3)

Proposition 0.21:

For square $\boldsymbol{A}$ ($m \times m$) the following hold:

1. $\mathrm{tr}(\boldsymbol{A}^\top) = \mathrm{tr}(\boldsymbol{A})$
2. $\det(\boldsymbol{A}^\top) = \det(\boldsymbol{A})$
3. $(\boldsymbol{A}^\top)^{-1} = (\boldsymbol{A}^{-1})^\top$, if $\boldsymbol{A}$ is nonsingular.
4. $\det(\boldsymbol{A}^{-1}) = 1/\det(\boldsymbol{A})$
5. λ is eigenvalue of $\boldsymbol{A}$ with eigenvector $\boldsymbol{v} \Rightarrow 1/\lambda$ is an eigenvalue of $\boldsymbol{A}^{-1}$ with eigenvector $\boldsymbol{v}$.
6. $\mathrm{rk}(\boldsymbol{A}) = m \Leftrightarrow \boldsymbol{A}^{-1}$ exists
7. $\mathrm{rk}(\boldsymbol{A}) = m \Rightarrow \boldsymbol{A}^+ = \boldsymbol{A}^{-1}$

Orthogonal Matrices

Proposition 0.22:

$\boldsymbol{A}$ is orthogonal $\Leftrightarrow$

1. $\boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I}_m \Leftrightarrow \boldsymbol{A}^\top\boldsymbol{A} = \boldsymbol{I}_m$.
2. $\boldsymbol{A}^\top$ is orthogonal.
3. $\boldsymbol{A}^{-1}$ is orthogonal.

Partitioned Inverses

Proposition 0.23:

$\boldsymbol{A}\ (m \times m), \boldsymbol{B}\ (m \times n), \boldsymbol{C}\ (n \times m), \boldsymbol{D}\ (n \times n)$:

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}^{-1} =$$

$$\begin{pmatrix} \boldsymbol{A}^{-1} + \boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & -\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}) \\ -(\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & (\boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1} \end{pmatrix},$$

provided the respective inverses exist.

Partitioned Inverses

Proposition 0.24:

For nonsingular $\boldsymbol{A}_i$ $(m_i \times m_i)$, $i = 1, \dots, r$:

$$\begin{pmatrix} \boldsymbol{A}_1 & & \mathbf{O} \\ & \ddots & \\ \mathbf{O} & & \boldsymbol{A}_r \end{pmatrix}^{-1} = \begin{pmatrix} \boldsymbol{A}_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \boldsymbol{A}_r^{-1} \end{pmatrix}$$

Relationship Between the Properties

Proposition 0.25:

1. $\boldsymbol{B}$ is nonsingular $\Rightarrow$ $\det(\boldsymbol{BAB}^{-1}) = \det(\boldsymbol{A})$
2. $\det(\boldsymbol{I}_m) = 1$
3. $\boldsymbol{A}$ $(m \times m)$ with eigenvalues $\lambda_1, \ldots, \lambda_m$: $\det(\boldsymbol{A}) = \prod_{j=1}^{m} \lambda_j$
4. $\boldsymbol{A}$ $(m \times m)$: $\mathrm{rk}(\boldsymbol{A}) < m \;\Leftrightarrow\; \det(\boldsymbol{A}) = 0$
5. $\mathrm{rk}(\boldsymbol{A}) = m \;\Leftrightarrow\; \det(\boldsymbol{A}) \neq 0$
6. $\boldsymbol{A}$ is singular $\Leftrightarrow$ $\det(\boldsymbol{A}) = 0$
7. If $\boldsymbol{A}$ and $\boldsymbol{D}$ are square, then
$$\begin{vmatrix} \boldsymbol{A} & \boldsymbol{B} \\ 0 & \boldsymbol{D} \end{vmatrix} = |\boldsymbol{A}||\boldsymbol{D}|$$
8. If $\boldsymbol{A}$ and $\boldsymbol{D}$ are square and $\boldsymbol{A}^{-1}$ exists, then
$$\begin{vmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{vmatrix} = |\boldsymbol{A}||\boldsymbol{D} - \boldsymbol{CA}^{-1}\boldsymbol{B}|$$

Relationship Between the Properties

Proposition 0.26:

For idempotent $\boldsymbol{A}$ ($m \times m$):

1. $\text{rk}(\boldsymbol{A}) = m \;\Rightarrow\; \boldsymbol{A} = \boldsymbol{I}_m$
2. $\text{rk}(\boldsymbol{A}) = \text{tr}(\boldsymbol{A})$
3. $\text{rk}(\boldsymbol{I}_m - \boldsymbol{A}) = m - \text{rk}(\boldsymbol{A})$

Relationship Between the Properties

Proposition 0.27:

For square $\boldsymbol{A}$ $(m \times m)$:

1. $\boldsymbol{A} = \text{diag}(a_{11}, \dots, a_{mm}) \Rightarrow a_{11}, \dots, a_{mm}$ are the eigenvalues of $\boldsymbol{A}$.
   In fact, triangularity of $\boldsymbol{A}$ suffices.
2. $\boldsymbol{A}$ is symmetric $\Rightarrow$ all eigenvalues of $\boldsymbol{A}$ are real numbers.
3. $\boldsymbol{A}$ is idempotent $\Rightarrow$ all eigenvalues of $\boldsymbol{A}$ are 0 or 1.
4. $\boldsymbol{A}$ is singular $\Leftrightarrow$ 0 is an eigenvalue of $\boldsymbol{A}$.
5. $\boldsymbol{A}$ is nonsingular $\Leftrightarrow$ all eigenvalues of $\boldsymbol{A}$ are nonzero.
6. $\boldsymbol{A}$ symmetric: $\boldsymbol{A}$ is positive definite $\Leftrightarrow$ all eigenvalues of $\boldsymbol{A}$ are real and greater than 0.
7. $\boldsymbol{A}$ symmetric: $\boldsymbol{A}$ is positive semidefinite $\Leftrightarrow$ all eigenvalues of $\boldsymbol{A}$ are real and greater than or equal to 0.
8. $\boldsymbol{A}$ has eigenvalues $\lambda_1, \dots, \lambda_m : \text{tr}(\boldsymbol{A}) = \sum_{j=1}^{m} \lambda_j$.

Relationship Between the Properties

Proposition 0.28:

$\boldsymbol{A}$ positive (semi) definite $\Leftrightarrow$

$$\det \begin{pmatrix} a_{11} & \cdots & a_{1i} \\ \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} \end{pmatrix} > 0 \ (\geqslant 0) \quad \text{for } i = 1, \ldots, m.$$

$\boldsymbol{A}$ negative (semi) definite $\Leftrightarrow$

$$\det \begin{pmatrix} a_{11} & \cdots & a_{1i} \\ \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ii} \end{pmatrix} \begin{cases} > 0 \ (\geqslant 0) & i \text{ even} \\ < 0 \ (\leqslant 0) & i \text{ odd} \end{cases} \quad \text{for } i = 1, \ldots, m.$$

Spectral Decomposition of a Symmetric Matrix

Theorem 0.29:

Let $\boldsymbol{A}$ $(m \times m)$ be symmetric with eigenvalues $\lambda_1, \ldots, \lambda_m$. Then,

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top,$$

where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$ and $\boldsymbol{U}$ is the real orthogonal $(m \times m)$ matrix whose columns are the orthonormal eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ of $\boldsymbol{A}$ (i.e. $\boldsymbol{v}_i^\top \boldsymbol{v}_j = 1$ if $i = j$ and 0 else) associated with $\lambda_1, \ldots, \lambda_m$. In other "words,''

$$\boldsymbol{A} = \sum_{j=1}^{m} \lambda_j \boldsymbol{v}_j \boldsymbol{v}_j^\top.$$

Cholesky Decomposition

Theorem 0.30:

$\boldsymbol{A}$ $(m \times m)$ symmetric positive definite:
There exists a unique lower (upper) triangular $(m \times m)$ matrix $\boldsymbol{B}$ with real positive diagonal elements such that $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{B}^\top$.

Example 0.31:

$\boldsymbol{A}$ (2×2)

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} b_{11} & 0 \\ b_{12} & b_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ 0 & b_{22} \end{pmatrix} \Rightarrow$$

$$\begin{aligned} b_{11} &= \sqrt{a_{11}}, \\ b_{12} &= a_{12}/b_{11}, \\ b_{22} &= \sqrt{a_{22} - b_{12}^2}. \end{aligned}$$

Let $f(\boldsymbol{x})$ be a differentiable real-valued function of the $(m \times 1)$ vector $\boldsymbol{x} = (x_1, \dots, x_m)^\top$.

Definition 0.32: Gradient.

The **gradient** of $f(\cdot)$ is

$$\frac{\partial f}{\partial \boldsymbol{x}} := \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_m} \end{pmatrix} \quad (m \times 1)$$

and

$$\frac{\partial f}{\partial \boldsymbol{x}^\top} := \left(\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_m} \right) \quad (1 \times m)$$

Definition 0.33: Hessian.

The **Hessian** of $f(\cdot)$ is

$$\frac{\partial^2 f}{\partial \boldsymbol{x} \partial \boldsymbol{x}^\top} := \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_m \partial x_m} \end{pmatrix} \quad (m \times m)$$

Definition 0.34: Jacobian.

Let $\boldsymbol{y}(\boldsymbol{x}) = (y_1(\boldsymbol{x}), \ldots, y_n(\boldsymbol{x}))^\top$ be an $(n \times 1)$ vector of differentiable functions of the $(m \times 1)$ vector $\boldsymbol{x} = (x_1, \ldots, x_m)^\top$. That is, $\boldsymbol{y}$ is a function mapping a subset of $\mathbb{R}^m$ on a subset of $\mathbb{R}^n$.

The **Jacobian** of $\boldsymbol{y}(\cdot)$ is defined as

$$\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}^\top} := \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \cdots & \frac{\partial y_n}{\partial x_m} \end{pmatrix} \quad (n \times m)$$

Matrix Calculus

Proposition 0.35:

We have

1. $\boldsymbol{x}$, $\boldsymbol{a}$ $(m \times 1)$

$$\frac{\partial \boldsymbol{a}^\top \boldsymbol{x}}{\partial \boldsymbol{x}} = \boldsymbol{a}$$

2. $\boldsymbol{x}$ $(m \times 1)$, $\boldsymbol{A}$ $(m \times m)$

$$\frac{\partial \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}}{\partial \boldsymbol{x}} = (\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{x}$$

   (Note how the formula simplifies if $\boldsymbol{A}$ is symmetric.)

3.
$$\frac{\partial^2 \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}}{\partial \boldsymbol{x} \partial \boldsymbol{x}^\top} = (\boldsymbol{A} + \boldsymbol{A}^\top)$$